Human-Aligned Chess AI: A Multitask Transformer for Humanlike Decision-Making

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***Abstract*—At the moment, it's clear that AI has surpassed human ability in almost every field. But, how useful really is this to us? Several fields (eg. law, education, games) have noticed that having a "perfect" AI isn't as good as it seems in all cases. One of the marquee examples is law, where an AI wouldn't consider highly important parts of human thinking such as fairness and ethics, it would sentence the victim based on raw logic, something that we clearly want to avoid. Being able to make AI that can understand humans and make decisions like them is becoming much more important in order to be able to make critical decisions that others can interpret and collaborate with.**

Being able to completely model human reasoning in any field is not possible due to the amount of 'noise' in each individual, but we aim for this in the game of chess. Chess is one of the most complex and popular games due to its large fan base and the vast amount of positions that can arise, despite the test of time chess remains unsolved. There are multiple existing chess engines, such as Stockfish and Alphazero, which are essentially 'perfect' forms of AI, they aim to be as strong as possible and win as many games as possible rather than trying to play like a human. We introduce Harmonia, an encoder-only Transformer trained through multitask learning to predict human chess moves and game outcomes. Although there have been multiple attempts to emulate human play in chess through the Maia project, Allie Chess, etc., our model is able to surpass all of these existing baselines despite being significantly smaller in size and trained on fewer data.

Although Harmonia is extremely good at modeling human play in chess, it is much more important to promote the development of human-AI interaction. It is unique in the sense that it was trained on human data in order to learn human policies and internal representations, which can be used as a tool for further research on educational uses and understanding how humans think. The results in this paper show that there is a high potential for artificial intelligence that can mimic human behavior to be more understandable and collaborative with fellow humans.

***Index Terms*—Artificial Intelligence, Human-AI Interaction, Chess, Transformer, Move Prediction, Multitask Learning, AI Alignment, Human Cognition**

I. Introduction

The main downside of having "perfect" AI systems is the lack of interpretability as to how these models came to their decisions. These models are essentially black boxes since we don't have the tools to understand how they came to their decisions, one of the biggest problems in chess. As mentioned above, there are several chess engines that are superhuman compared to us, but they play in a way that is very peculiar to humans because they make decisions based on brute force search. Though these engines have revolutionized the field of chess because they are extremely powerful preparation tools, it's difficult to cooperate with them for educational purposes to understand why they came to their decisions.

Previous works have attempted to match human play through simply weakening the AI, such as by reducing the amount of roll-outs of search that it can perform. Although this is an improvement compared to the original AI because this new, weaker AI is now beatable by humans, it still hasn't captured the subtle decisions that humans make in chess. This is something that is generalizable to all fields, simply weakening an AI won't make it "human" because it won't be able to capture the human nuances in decision making.

In chess, each player starts of with a set amount of time to complete the game, if they run out of time then they lose the game. To date, no AI system considers the remaining amount of time the player had when making the move, despite this being a crucial factor in determining the quality of the move that the player would play. The less time the player has, the less time they have to calculate the position deeply which would decrease the quality of their moves. Especially in the fast paced world that we live in today, where the majority of games are played at fast time controls (both players start with a low amount of time), the amount of time that the player had remaining when making the move is a key factor when predicting the move that the player will make. Harmonia's ability to take into account the amount of time the player had when making the move is the biggest differentiating factor from all other existing engines.

II. Related Works

There have been multiple approaches to represent chess as a model system, starting with Deep Blue [7] in the 1990s, which used brute force search to outplay humans. Recently, machine learning techniques such as reinforcement learning have come to the forefront of engine development in chess, as can be seen by AlphaZero [8] and Leela Chess Zero (LC0). These engines have been trained purely on self play, making them very different to how humans think and understand chess. Recently, transformers have been making waves in development toward more human like systems in both chess and other fields, as

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can be seen by the Chess Transformers [6] library which is a family of transformers from derived from Attention is All You Need [1]. Harmonia is an extension of the encoder-only transformer in the Chess Transformers [6] library.

More recently, the Maia project [3] enjoyed success in using the popular ResNet framework to create a model that can closely match human behavior in chess. Although it has a strong predictive performance, it requires move history in order to predict the next move and this is a significant drawback because the model won't be reliable in positions where move history isn't available. Harmonia avoids this problem by only focusing on the current board state because in a perfect world, a human should be able to make a moves from any given position without the move history.

Previous work has also utilized sequential data in an attempt to predict game outcomes in chess, as seen in [4]. In this context, an LSTM was used in an attempt to capture the previous moves to a chess position but with little accuracy when compared to other approaches. Harmonia builds on this by combining outcome prediction along with the move prediction task in a single model, we found that the Harmonia's move prediction actually improved simultaneously with the outcome prediction accuracy.

III. Methodology

A. Board Position Representation

Transformers require data that is represented sequentially. Since, the most important feature that our model needs in order to predict the next move is the board position, we want to make sure that our encoding of the board position keeps as much information as possible. A chess board is an 8 by 8 matrix (64 squares), each square alternates between the colors white and black; in order to encode this we flatten this 8 by 8 matrix into a vector of length 64. Each element in the vector corresponds to a square on the chess board, if the square doesn't have any piece on it then that element in the vector has a value of 0. Likewise, each chess piece is assigned a different integer value from 2 to 13 because there are 12 different piece types (6 for each color). If there is an en passant square then we encode that square with a value of 1. Castling right is also a key element of the board position, we use 4 boolean vectors to represent the castling rights for each player and another boolean vector which represents which player's turn it is.

B. Auxiliary Information

In chess, there are several auxiliary features that humans consider when pondering a particular chess position. To improve the model's understanding of human like chess, we incorporate these features into the input so that the model has additional context to understand more strategic and nuanced aspects of the game. One of the key features that we incorporate is the game phase, in chess there are generally 3 phases of the game (opening, middlegame, and endgame), we represent these 3 phases with a one-hot vector with values (0, 1, 2) corresponding with each of these phases of the game. We decided that if the position had more than 26 pieces on the board then the game was considered to be in the opening phase, if it had between 26 and 14 pieces then it was in the middlegame, and if both of these conditions aren't satisfied then we say the position is in the endgame phase. These are arbitrary values for separating the 3 phases of the game because there is no set definition for these phases. But, there is a general consensus within the chess community that the opening phase is when most of the pieces are still present on the board, the middlegame is when few pieces have been exchanged, and the endgame is when the position has greatly simplified with few pieces remaining.

In chess, the term time control refers to the amount of time that each player starts with, there is a form of time control that is referred to as increment: after making a move the player gains n seconds. The time control significantly influences how humans play because with less time the quality of the moves that are made decreases, we represent the base amount of time and the increment time as separate continuous, linear vectors in the embedding space. We encode the amount of time that the player had in the current position in the same way. We also encode other auxiliary attributes that we thought were important for humans when considering which move to make:

Move number, which helps the model track game progression. Number of legal moves, which tells the model how complex a position is. Material balance, we calculate the total material value for white and black separately. Material difference from the perspective of the player to move, calculated as (White's material Black's material) if it is White's turn, or (Black's material White's material) if it is Black's turn.

C. Model Architecture

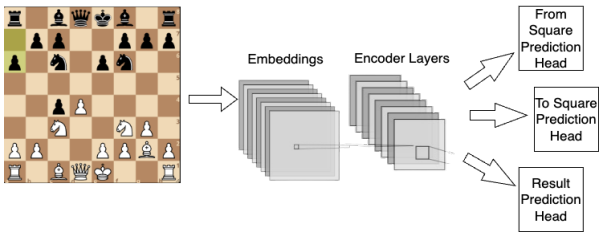

Fig. 1. Overview of Harmonia's architecture.

Harmonia is an encoder-only transformer, derived from the original Attention is All You Need paper [1]. Each input vector for the model, discussed above, is projected into an embedding space and then processed through an encoder. The encoder consists of a stack of transformer encoder layers, each comprising:

- Multi-Head Self-Attention to capture long-range dependencies between board features.
- Position-Wise Feed-Forward Networks (FFN) for non-linear transformations.
- Layer Normalization and Dropout to enhance stability and generalization.

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To facilitate multi-task learning, the model incorporates two CLS tokens [10], each corresponding to a distinct prediction head:

- Game result `[CLS]` token – used for predicting the game outcome.

`[CLS]` tokens have been shown to be helpful for classification tasks and text generation, which is why we decided to use them. Each `[CLS]` essentially aggregates the necessary information into one vector that we can find the necessary logits over. Two linear prediction heads find logits over the 64 board squares outputted by the encoder, where each square is competing to be labeled as either the 'from' or 'to square' of the move. Another linear prediction head finds logits over the corresponding game result `[CLS]` token to predict the outcome of the game.

D. Hyperparameters

During training we used a batch size of 512, and we trained our model on an nvidia A100 GPU for around 24 hours. We accumulated gradients for 4 steps before going through a backward pass, in order to simulate a larger batch size. We used a embedding size of 512, with 8 heads in our multihead attention. Also, we had a dropout of 0.1 and 6 hidden layers in our encoder. In total, we trained for 100000 steps, with the learning rate increasing linearly up to 0.001 for the first 3000 steps and then decaying exponentially for the rest of the steps.

E. Dataset

Harmonia was trained on a dataset constructed from a large open source database of human chess games, lichess.org. In order to compare our results with the current leading baseline in this field, Maia [3], we trained our model on the same dataset as theirs, except that we only used players with a rating between 1900 and 2000. We chose this range because this range had the most number of games played, which meant that there would be an abundance of data to train our model on.

IV. Results

A. Move Prediction

In order to evaluate our model on how "human" it is, the main criteria is how often it matches the moves that a human would play in a given position. We call this criteria move-matching accuracy, as was used to evaluate Maia2, and stockfish. The table below shows Harmonia's move matching accuracy compared to these other engines. All evaluations of our model were conducted on our predefined test set, the same test set that Maia was evaluated on [3]. The test contains approximately 200,000 that were played on lichess.org during December of 2019, none of the test set was used as training data for the model.

Clearly Harmonia is able to outperform both Maia2 and Stockfish, but we see the biggest gain in performance when we test Harmonia on low time positions. We categorize low time positions as a position where either player has less than 30 seconds to make all their moves to complete the game. This large performance gain is likely because Harmonia included low time positions in its dataset while a model like Maia2 didn't. Also, this metric shows the usefulness of having two embeddings that capture the remaining amount of time that each player has when making the move.

	Harmonia	Maia2	Stockfish
No Low Time Positions	0.5266	0.5121	0.412
Low Time Positions	0.5334	0.5269	0.402

TABLE I
HARMONIA'S MOVE MATCHING ACCURACY COMPARED TO OTHER MODELS

Each move in a chess game is associated with a move number, since chess moves are sequential the later moves of the game have a larger move number than the earlier ones. Fig 2. shows how Harmonia's accuracy varies as a game goes on. The accuracy varies more at the start due to the large number of opening positions that are possible, but as the time goes on the accuracy increases, showing that Harmonia has learned a deep understanding of human gameplay in the middle (moves 15-25) and endgame (moves 25-40). When comparing Maia's graph in Fig 3., we see a stark difference. Maia's graph starts out low, once again due to the large number of opening positions and because Maia explicitly wasn't trained on any opening positions, but it stays relatively the same throughout rest of the game. This shows that Maia hasn't developed as deep of an understanding of human gameplay because it's accuracy in the middlegame (moves 15-25) aren't as high Harmonia's.

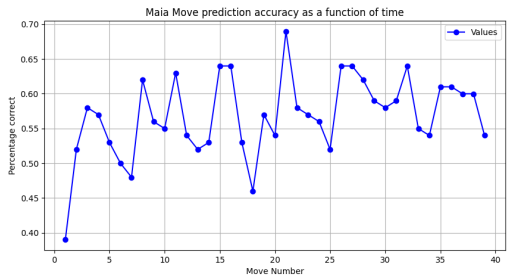

Fig. 2. Maia's accuracy in relation to move number

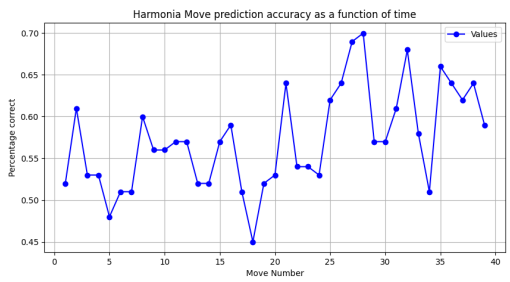

Fig. 3. Harmonia's accuracy in relation to move number

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B. Game Result Prediction

There are three possible outcomes of a chess game: win, loss, or draw. So, if the model were simply guessing, it would get an accuracy of around 33% by random chance. But, since the model achieves an accuracy of around 64%, this shows that it has practically doubled the random chance accuracy, meaning that it has learned an efficient internal representation to determine the outcome of a game from any given position. Our game result prediction head significantly outperforms all previous works by around 15% [5]. Previous works like Maia [3] and Allie chess [9] have used value heads that output values between -1 and 1, where 1 represents a white win, -1 represents a black win, and 0 represents a draw. We experimented with this approach, but found that with the lack of label smoothing present, after many steps of training all values tended to either go to one of the extremes and didn't give fair values for each outcome, eg. a slightly better position for one sided would be given an extreme value (-1 or 1) rather than a more appropriate evaluation like 0.35 or -0.43. Our approach uses label smoothing to output proper probabilities for each outcome.

C. Learning Rules of Chess

Our model outperforms state of the art benchmarks in the field, but has it actually learned to play chess? We evaluate this criteria by saying how often the model plays legal moves in different game states. In chess there are only a certain amount of legal moves in a given position, an illegal move won't be allowed to be played. We use the same criteria that was used in Allie Chess [9].

TABLE II
EVALUATION OF MOVE VALIDITY

Evaluation set	Top-1 move is valid (%)
Lichess	100.0 ± 0.0
Lichess (under *check*)	99.4 ± 0.0
Random	97.2 ± 0.0
Random (under *check*)	95.1 ± 0.0

D. Ablation Study

While coming up with the model architecture for Harmonia, there were multiple choices for auxiliary information that we considered. There were four possible tasks that we were considering for our model to predict: the next move, the outcome of the game, the number of moves until the game ends, and the amount of time the model should take to play the move. We decided, that we should make the model output the outcome of the game and the next move in all our ablation tests because these were the most important tasks to compare with baseline models. Each ablation is a combination of these 4 tasks, and how it affects the move prediction accuracy and the game outcome prediction accuracy. We conducted the ablation tests on 10% of our full dataset.

- **ablation 1**: Model only outputs move prediction and game outcome prediction.
- **ablation 2**: Model outputs move prediction, game outcome prediction, and predicts the number of moves until the game ends.
- **ablation 3**: Model outputs move prediction, game outcome prediction, predicts number of moves until the game ends, and predicts the amount of time to spend 'pondering' the position.
- **ablation 4**: Model outputs move prediction, game outcome prediction, and predicts the amount of time to spend 'pondering' the position.

	ablation 1	**ablation 2**	**ablation 3**	**ablation 4**
Move Prediction	0.508	0.507	0.498	0.504
Game Outcome	0.620	0.612	0.586	0.605

TABLE III
ABLATION TEST RESULTS

As a result of these ablation tests, it's clear that the model performed best when it only outputted a move prediction and a game outcome prediction. Since, we didn't apply any normalization to the move time predictions and moves until end predictions, these tasks were inherently noisy which is why they may have significantly affected the move prediction task. The clear trend is that the fewer tasks that the model predicts, the better that it performs.

V. DISCUSSION

In this paper, we show how Transformers can significantly outperform other architectures when modeling human decision making, despite having a significantly smaller model and less data than others. We think that this work could have a large impact on other works where modeling human behavior is crucial for success. Especially, in education settings where human like AI is increasingly being looked for, the techniques discussed in this paper have a large potential to be used there. It would be interesting to see if we could incorporate XAI (explanable AI) into our methodology, so that our model could explain the reasoning behind each move that it makes. Also, as more works like these where models are created which mimic human behavior, it will be more and more important for proper Turing tests to be developed to see how human these models really are. The creation of Turing tests to see if people can see the difference between these human like chess models like Maia, Harmonia, Allie, etc. would be interesting to explore.

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